Monte Carlo and Finite Difference Methods for Option Pricing: A Numerical Investigation

1 Questions and experimental design

The two contracts create different numerical difficulties. The digital payoff jumps at the strike, whereas the barrier call can be knocked out between simulated dates. I compare Monte Carlo (MC) and finite differences by separating sampling error from discretisation effects. The risk-neutral model and parameters are

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t, \qquad (S_0, r, \sigma, T) = (100, 0.05, 0.25, 1).$$

For Extension B, I test whether a geometric control variate makes arithmetic Asian pricing more efficient once its extra simulation cost is included.

Problem	**Methods used**	**Investigation**	**Main conclusion**
Digital	Standard, antithetic and importance-sampled MC; explicit, implicit and CN finite differences (FD)	Equal simulation budgets; payoff averaging; grid convergence and stability	Importance sampling improves MC at $K = 140$. Averaged CN with implicit startup gives much smaller errors at comparable cost.
Barrier	Discrete and bridge MC; explicit, implicit and CN FD	Missed crossings; a boundary at the barrier; accuracy versus runtime	Discrete monitoring overprices. Use bridge conditioning for simulation and damped CN for high precision.
Asian (B)	Standard MC and a geometric control variate	Variance reduction, independent validation and cost	At one million paths, the control reduces variance about 835-fold. The gain remains substantial after allowing for pilot cost.

Design and reproducibility. MC samples are generated from the model, and FD prices are computed on a grid. The accompanying Python program saves the results, seeds and figures. It uses Python 3.13.7, NumPy 2.3.3 and SciPy 1.16.2 on Windows, with base seed 20260914 and separate experiment offsets. Reported runtimes are medians of three complete calls, excluding plotting and file output; Asian control timings include the independent pilot. They measure this implementation on this machine.

For independent discounted observations Y_i, I report their mean, standard error $\mathrm{SE} = s_Y/\sqrt{n}$ and approximate 95% interval $\overline{Y} \pm 1.96\mathrm{SE}$. An antithetic pair is one observation. FD error is measured against a known or independently evaluated benchmark. Replicated MC runs avoid judging accuracy from one fortunate estimate; successive grid refinements check whether FD results have settled.

2 Digital call: simulation and variance reduction

The digital pays $\mathbf{1}_{\{S_T>140\}}$. Write $\mu = r - \sigma^2/2$ and $a = [\log(K/S_0) - \mu T]/(\sigma\sqrt{T})$. Exact terminal simulation uses

$$S_T = S_0 \exp(\mu T + \sigma\sqrt{T}Z), \quad Z \sim N(0,1), \qquad V_D = e^{-rT}\Phi(-a) = 0.0969151345.$$

Here Φ is the standard normal distribution function. The exercise probability is only $p = \Phi(-a) \simeq 0.1019$, so most standard samples contribute zero. There is no time discretisation error because only the terminal stock price is needed.

I compare the sampling methods at a fixed payoff budget. Antithetic MC averages the payoffs obtained from Z and $-Z$. For importance sampling (IS), draw $Z \sim N(0,1)$ and shift to $X = Z + a$. The observation is

$$Y_{\mathrm{IS}} = e^{-rT}\mathbf{1}_{\{X>a\}} \exp(-aZ - a^2/2).$$

The exponential is the original normal density divided by the shifted density, so the expectation is unchanged. The threshold lies at the new mean and half the samples exercise. This simple shift is held fixed as a rule; it is not claimed to be optimal.

Method	Price	SE	95% half-width	Time (s)
Standard	0.096970	0.000288	0.000564	0.0252
Antithetic	0.097482	0.000272	0.000532	0.0222
IS	0.096795	0.000125	0.000245	0.0516

Table 1: 10^6 payoff evaluations; antithetic uses 5×10^5 pairs.

The estimated equal-budget variance reductions are 1.12 for antithetic and 5.28 for IS. Counting pairs correctly matters: comparing the observation variances alone would overstate the antithetic gain by a factor of two. IS costs more per sample but reduces SE^2 $\times$ runtime by 2.57 relative to standard MC.

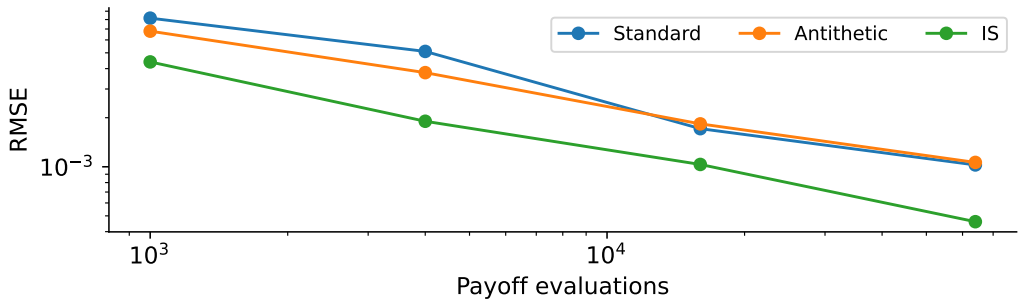


Figure 1: RMSE from 40 independent runs at each simulation budget.

Budgets $10^3, 4\times10^3, 1.6\times10^4, 6.4\times10^4$ give log–log slopes $-0.53, -0.45, -0.53$ for standard, antithetic and IS respectively. The results are consistent with $n^{-1/2}$ convergence: IS improves the constant, not the rate. The standard estimate in Table 1 happens to be closer to the exact price than IS; the SEs and repeated runs show why this is not evidence that standard MC is better. The antithetic interval misses the benchmark in this run, which can occur with a nominal 95% interval.

3 Finite differences: the digital discontinuity

With $u(\tau, S) = V(T - \tau, S)$, the pricing equation is $u_\tau = \frac{1}{2}\sigma^2 S^2 u_{SS} + rSu_S - ru$. On $S_j = j\Delta S$, central differences give

$$(L_h u)_j = a_j u_{j-1} + b_j u_j + c_j u_{j+1}, \qquad \begin{cases} a_j = (\sigma^2 j^2 - rj)/2, \\ b_j = -\sigma^2 j^2 - r, \\ c_j = (\sigma^2 j^2 + rj)/2. \end{cases}$$

The time update, with boundary contributions at the appropriate time levels, is

$$(I - \theta\Delta\tau L_h)u^{n+1} = (I + (1-\theta)\Delta\tau L_h)u^n.$$

Explicit Euler, implicit Euler and Crank–Nicolson (CN) use $\theta = 0, 1, 1/2$. I also test Rannacher-started CN (R-CN): replace the first full step by two implicit half-steps, then use CN [1]. Each implicit update is a tridiagonal solve.

To test how the discontinuity affects grid convergence, I use $S \in [0, 400]$, with $u(\tau, 0) = 0$ and $u(\tau, 400) = e^{-r\tau}$. The tested grids include both S_0 and K. Pointwise terminal data assign zero at K; cell averaging assigns $1/2$ there. Although changing one point does not affect the continuous expectation, a grid node has a finite numerical weight.

$M = N$	Raw R-CN error	Averaged error	Implicit error	R-CN (s)
100	9.60×10^{-3}	3.80×10^{-5}	3.37×10^{-4}	0.008
200	4.82×10^{-3}	9.36×10^{-6}	1.81×10^{-4}	0.015
400	2.41×10^{-3}	2.33×10^{-6}	9.37×10^{-5}	0.038
800	1.21×10^{-3}	5.82×10^{-7}	4.76×10^{-5}	0.076
1600	6.04×10^{-4}	1.46×10^{-7}	2.40×10^{-5}	0.222

Table 2: M spatial intervals and N time steps; implicit data are averaged.

Raw-payoff errors approximately halve, whereas averaged R-CN errors fall by four per doubling. Implicit Euler is approximately first order along $N = M$ because of its time error. The finest averaged price is 0.096915280. Keeping $\Delta S = 0.5$ and $N = 800$, increasing the upper boundary to 600 and 800 changes the price by less than 10^{-12}. Domain truncation therefore does not explain this refinement pattern.

Stability and cost. Here $a_j, c_j \geq 0$, and explicit Euler is monotone if $\Delta\tau[\sigma^2(M-1)^2 + r] \leq 1$. At $M = 400$, $N = 12000$ gives ratio 0.829 and a bounded solution; $N = 9000$ gives 1.106 and values exceeding 10^{100}. Using $N = \lceil T[\sigma^2(M-1)^2 + r]/0.9 \rceil = 11056$ leaves a safety margin and gives error 5.94×10^{-6} in 0.485 s. R-CN uses only 400 steps, giving 2.33×10^{-6} in 0.038 s.

With $M = 400, N = 20$, plain CN has two local decreases near the strike after its first step; R-CN removes both. Stability alone does not prevent oscillations. I recommend averaged R-CN for numerical precision: at $M = 200$ its error is already 9.36×10^{-6} in 0.015 s, below the million-sample IS standard error. For this particular contract, the closed-form price remains the simplest choice.

4 Barrier call: missed crossings in Monte Carlo

The up-and-out call has $K = 100$, $H = 130$ and discounted payoff

$$e^{-rT}(S_T - K)^+ \mathbf{1}_{\{\max_{0\le t\le T} S_t < H\}}.$$

Checking the barrier only at simulated endpoints accepts some paths that crossed H between dates. With a nonnegative payoff this creates upward bias.

I use exact lognormal increments to isolate missed crossings from Euler or Milstein path discretisation error. For log endpoints $x_i, x_{i+1} < h = \log H$, the conditional survival probability is [2]

$$q_i = 1 - \exp\left(-\frac{2(h - x_i)(h - x_{i+1})}{\sigma^2 \Delta t}\right), \qquad Y_B = e^{-rT}(S_T - K)^+ \prod_i q_i.$$

Set $q_i = 0$ if either endpoint reaches the barrier. Under the specified constant coefficients, this Brownian bridge weighting removes monitoring bias exactly. The payoff otherwise depends only on S_T, so one interval across $[0, T]$ suffices.

For an independent benchmark, put $x = \log S_0$, $v = \sigma^2 T$, and let ϕ_v denote the centred normal density with variance v. The reflection principle gives the density of the log price before hitting h:

$$f_{\text{kill}}(y) = \phi_v(y - x - \mu T) - e^{2\mu(h-x)/\sigma^2}\phi_v(y + x - 2h - \mu T).$$

Then $V_B = e^{-rT}\int_{\log K}^{h}(e^y - K) f_{\text{kill}}(y)\, dy = 2.2235389914$. Numerical quadrature with absolute and relative tolerances 10^{-12} agrees with direct integration of the bridge-weighted payoff within 10^{-11}. This provides a check separate from the FD solver.

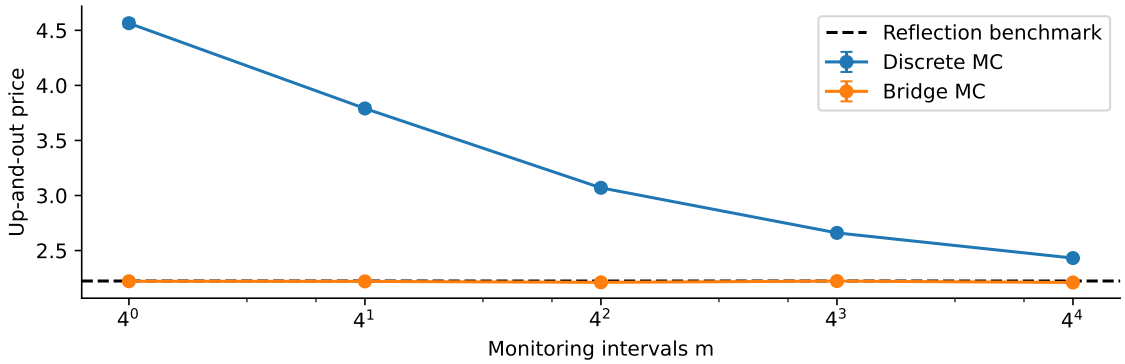


Figure 2: 200,000 paths per resolution; bars show 95% sampling intervals. Discrete and bridge estimators use the same increments at each resolution.

At 256 intervals, discrete monitoring gives 2.431888, an error of 0.208349, against SE 0.012288. The paired discrete-minus-bridge gap is 0.221800 with SE 0.004065, so sampling variation cannot plausibly explain it. The gap falls from 0.8585 to 0.4377 to 0.2218 at 16, 64 and 256 intervals: approximately halving when the step size quarters, consistent with a square-root monitoring error. More paths alone cannot remove this bias.

All five bridge intervals contain the benchmark. Their fluctuations reflect sampling error, without the systematic downward trend of discrete monitoring. A narrow MC interval is therefore insufficient evidence of accuracy unless monitoring bias is also controlled.

5 Barrier efficiency and the Asian extension

Barrier FD. The grid ends at H, with $u(\tau, 0) = u(\tau, H) = 0$ and terminal data $(S-K)^+$ for $S < H$. The absorbing boundary kills the option as soon as the stock reaches H. Each grid includes S_0, K, H. The terminal payoff jumps from almost 30 to zero at H, which can cause startup oscillations.

$M = N$	R-CN price	Absolute error	Time (s)	CN minimum
130	2.2218891	1.65×10^{-3}	0.009	-1.11
260	2.2231267	4.12×10^{-4}	0.022	-7.21
520	2.2234359	1.03×10^{-4}	0.045	-12.58
1040	2.2235132	2.58×10^{-5}	0.096	-16.98

Table 3: CN minimum is taken over every grid node and time level.

R-CN stays nonnegative and its error falls by about four per grid doubling. Plain CN has similar final accuracy, but the negative intermediate prices in Table 3 are unrealistic. At $M = 520$, explicit Euler takes 0.701 s and 18,706 steps for error 1.83×10^{-4}; R-CN takes 0.045 s and 520 steps for 1.03×10^{-4}. The implicit startup therefore improves the solution at little cost.

One-step bridge MC with 10^6 paths gives 2.221161 (SE 0.00342) in 0.142 s. R-CN at $M = N = 1040$ takes 0.096 s for error 2.58×10^{-5}, so I prefer FD for precision here. For simulation, one bridge interval is enough: at 200,000 paths, using 256 intervals instead of one raises runtime from 0.025 to 6.179 s and payoff variance from 11.69 to 25.68. Conditioning only on S_T averages out the extra randomness in intermediate states, explaining the lower variance.

Extension B: a geometric control for the Asian call. The contract has $K = 100$ and $m = 12$ monthly observations. To state the formulas generally, take m equally spaced dates $t_i = iT/m$, $i = 1, \dots, m$, and define $A = m^{-1} \sum_i S_{t_i}$ and $G = (\prod_i S_{t_i})^{1/m}$. Neither average includes S_0. The discounted payoffs are $X = e^{-rT}(A - K)^+$ and $C = e^{-rT}(G-K)^+$. I use C as a control because the two averages respond similarly to a stock path and $\mathbb{E}[C]$ is known [3].

$\log G$ averages Gaussian log prices, so it is Gaussian. Using $\mathrm{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j)$ gives its mean and variance:

$$m_G = \log S_0 + \mu T \frac{m+1}{2m},$$
$$v_G = \frac{\sigma^2}{m^2} \sum_{i=1}^{m} \sum_{j=1}^{m} \min(t_i, t_j) = \sigma^2 T \frac{(m+1)(2m+1)}{6m^2}.$$

The last equality uses $t_i = iT/m$ and $\sum_{i,j} \min(i, j) = m(m+1)(2m+1)/6$. Setting $m = 12$ gives $v_G = 0.0235098380$; the factors 13 and 25 are simply $m + 1$ and $2m + 1$. With $d_2 = (m_G - \log K)/\sqrt{v_G}$ and $d_1 = d_2 + \sqrt{v_G}$,

$$c_G = \mathbb{E}[C] = e^{-rT}\{e^{m_G + v_G/2}\Phi(d_1) - K\Phi(d_2)\} = 6.99073142.$$

The controlled payoff is $Y = X - \widehat{\beta}(C - c_G)$. I estimate $\widehat{\beta} = \widehat{\mathrm{Cov}}(X, C)/s_C^2 = 1.039867$ from 20,000 independent pilot paths and keep it fixed for all three production budgets. Independence gives $\mathbb{E}[Y \mid \widehat{\beta}] = \mathbb{E}[X]$. Exact simulation at the monthly dates avoids time discretisation bias.

6 Asian results and conclusions

Paths	Method	Price	SE	Time (s)	VR factor
10,000	Standard	7.255985	0.106835	0.005	1.0
10,000	Geo. CV	7.308936	0.003503	0.016	929.9
100,000	Standard	7.278630	0.033557	0.048	1.0
100,000	Geo. CV	7.310995	0.001155	0.054	843.8
1,000,000	Standard	7.302458	0.010629	0.565	1.0
1,000,000	Geo. CV	7.312727	0.000368	0.699	835.0

Table 4: $\widehat{\mathrm{VR}} = s_X^2/s_Y^2$ at equal production counts. CV timings include the 20,000-path pilot; both methods use the same production paths.

The controlled price at 10^6 paths is 7.312727, with 95% interval $[7.312006, 7.313448]$. An independent joint-normal simulation using Cholesky factorisation gives 7.312286 (SE 0.000260) from two million paths. The estimates differ by less than one combined SE; this is a statistical check, not an exact benchmark.

The decline in the estimated VR factor from 929.9 to 835.0 needs some care. With the same fixed $\widehat{\beta}$, both estimators' variances scale as $1/n$, so their population ratio does not depend on n. Table 4 instead uses sample variances. From 10^4 to 10^6 paths, s_X^2 changes from 114.136 to 112.984, while s_Y^2 rises from 0.12274 to 0.13531. The smaller run therefore gives a higher ratio, mainly because it estimates a smaller residual variance. The runs use different production seeds. These three values do not establish a downward trend in the method's effectiveness; even at 10^6 paths, SE is about 29 times smaller.

Including pilot time, the variance–runtime gain is about 278 at 10^4 paths and 675 at 10^6: the fixed pilot cost matters less in larger runs. Across 80 further runs with new pilots, the controlled-price standard deviation is 0.00372, close to the RMS reported SE of 0.00367.

Why FD becomes harder. The Asian value depends on S and the accumulated sum B. Between observation dates, it satisfies the Black–Scholes PDE in S with B fixed. At each date, $V(t_i^-, S, B) = V(t_i^+, S, B + S)$, and after the final update the payoff is $(B/12 - 100)^+$. The extra sum grid and interpolation increase FD work from $O(M_S N_t)$ to $O(M_S M_B N_t)$. MC needs only the 12 monthly increments and running sums.

I therefore prefer exact-date MC with the geometric control for the Asian call. It handles the average directly and gives a large precision gain after pilot cost, while the one-state FD method is more efficient for the core problems tested here.

References

[1] M. B. Giles and R. Carter (2006). Convergence analysis of Crank–Nicolson and Rannacher time-marching. *Journal of Computational Finance*, 9(4), 89–112.

[2] P. V. Shevchenko and P. Del Moral (2015). Valuation of barrier options using sequential Monte Carlo. arXiv:1405.5294, revised version.

[3] P. Boyle, M. Broadie and P. Glasserman (2001). Monte Carlo methods for security pricing. In *Option Pricing, Interest Rates and Risk Management*, pp. 185–239.